9C Lesson 5

Medians and quartiles

Year 9 Science · 9C Threat from disease · Student Book pages 46 to 47

Name _____ Class _____ Date _____

Today I can:

☐ I can find the median, the lower and upper quartiles and the interquartile range of a simple data set.

☐ I can use quartiles to compare the variation in two groups and to interpret the results of a medicine trial.

Key words

median the middle value when the values are written in order

mean the total of the values divided by the number of values

range the largest value minus the smallest value

quartile a value one quarter or three quarters of the way through the ordered data

lower quartile the value one quarter of the way through the ordered data (LQ)

upper quartile the value three quarters of the way through the ordered data (UQ)

interquartile range UQ minus LQ: the spread of the middle 50% of the data

normal distribution the bell-shaped curve seen when a continuous variable is plotted for a population

continuous variation variation that can take any value in a range, e.g. height

clinical trial a test of a medicine on groups of people

● Core Everyone does this section.

1. Nine ladybirds were counted for spots: 7, 2, 9, 4, 7, 5, 2, 8, 6.

a) Write the values in order, smallest first.

b) State the median.

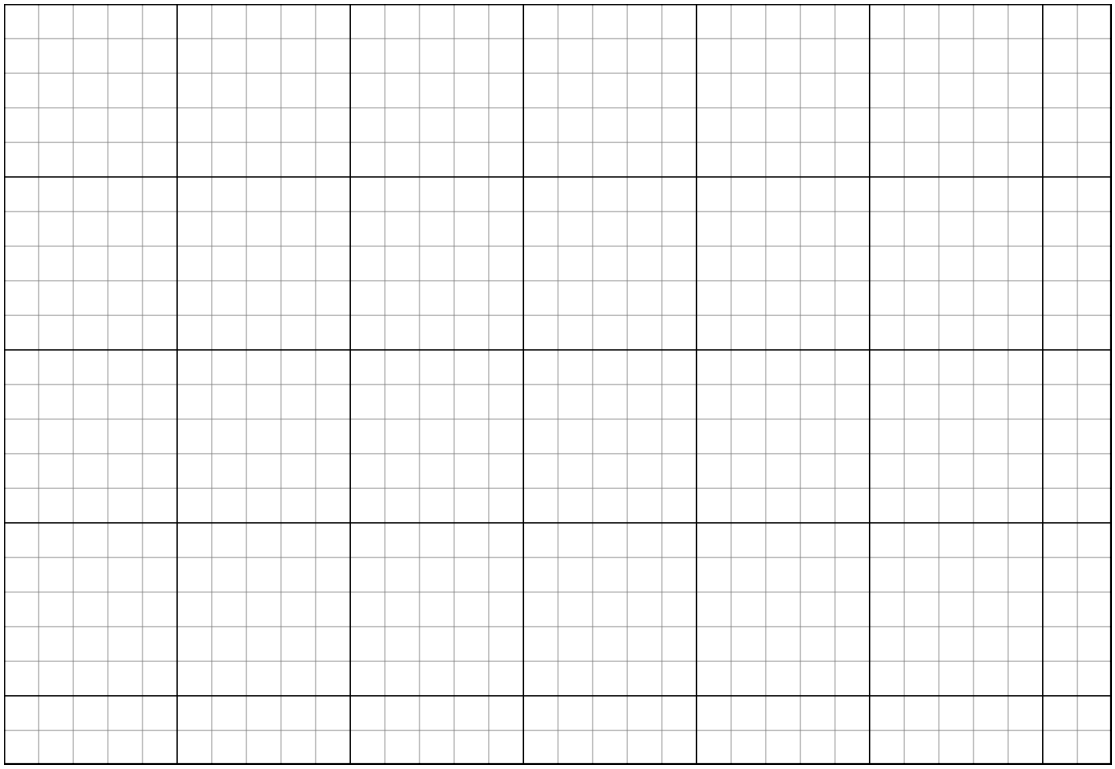
c) Calculate the range.

2. The foot lengths (cm) of 36 male students were: 25.7, 29.1, 26.4, 26.5, 28.2, 27.9, 27.9, 28.1, 26.9, 27.3, 27.2, 28.8, 27.9, 27.4, 28.2, 26.6, 27.3, 27.8, 26.3, 27.3, 28.0, 26.6, 27.1, 26.6, 26.4, 25.9, 27.3, 25.1, 28.5, 26.7, 28.8, 26.6, 28.6, 29.0, 25.7, 28.9.

a) Complete the table by counting the number of students in each group. Tick off each value as you use it.

Foot length (cm)	**25.1 to 25.5**	**25.6 to 26.0**	**26.1 to 26.5**	**26.6 to 27.0**	**27.1 to 27.5**	**27.6 to 28.0**	**28.1 to 28.5**	**28.6 to 29.0**	**29.1 to 29.5**
Number of students									

b) Plot a bar chart of the grouped data on the grid. The bars should touch. Label both axes.



c) Identify the group that contains the median foot length.

d) Does your bar chart show a normal distribution? Explain your answer.

●● Stretch When Core is finished.

S1. Here is a data set: 28, 13, 4, 25, 21, 14, 6, 19, 35, 17, 22, 34, 11, 16, 8.

a) Write the values in order.

b) Find the median value.

c) Find the lower quartile and the upper quartile.

d) Calculate the interquartile range.

S2. Class data: copy the class hand span values in order. Then find the median, the quartiles and the interquartile range.

●●● Challenge Explain in full sentences.

C1. In a statin trial, patients were measured for their ability to absorb cholesterol and split into quartiles. The table shows the change in their risk of a heart attack after taking statins.

Quartile of cholesterol absorption	lowest	lower middle	upper middle	top
Change in risk of heart attack (%)	-37.7	-34.3	-24.7	+16.6

a) Explain why the data for cholesterol absorption were split into quartiles.

b) The quartile boundaries of the cholesterol absorption index were 107, 126 and 148. Calculate the interquartile range.

c) Explain what the results show about the effect of cholesterol absorption on how well statins reduce the risk of a heart attack.

C2. Explain why the interquartile range is more useful than the range for comparing the variation in two different groups.

My thinking (any language)

A newspaper says 'the average family has 1.7 children'. Which kind of average is this, and why can no family have 1.7 children? When would the median be more useful than the mean?

Word bank median · mean · mode · range · lower quartile · upper quartile · interquartile range · normal distribution · order · clinical trial

✂ ---------- cut here

Exit ticket 9C Lesson 5: Medians and quartiles Name _____

1. Write these values in order and give the median: 12, 5, 9, 20, 7.

2. The upper quartile is 30 and the lower quartile is 18. Calculate the interquartile range.

3. What does a small interquartile range tell you about a population?

Today I can: Goal 1 ☐ Goal 2 ☐